INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

SWEATPANTS, PHYSICAL FITNESS UNIFORM (PFU)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Departments of Defense.

1. SCOPE

1.1 Scope. This specification covers a knitted jersey fleece sweatpant used as a component of the Physical Fitness Uniform.

1.2 Classification. The sweatpants shall be of one type in the following sizes (see 6.2).

SIZES

X-Small		Large
Small	Medium	X-Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-E-20652	-	Eyelet, Metallic, and Eyelet Washers, Metallic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment, Equipage; General Specification For
MIL-T-43548	-	Thread, Polyester, Cotton-Covered and Rayon Covered
MIL-T-43624	-	Thread, Polyester, Spun

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets
MIL-STD-1491	-	Glossary of Knitting Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Flammable Fabrics Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Chromatic Transference Scale

Dimensional Changes in Automatic Home Laundering of Garments

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215,Research Triangle Park, NC 27709.)

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standard Thread Shades

(Application for copies should be addressed to the Color Association of the United States, Inc., 343 Lexington Ave., New York, NY 10016. If color samples are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.2 Samples.

3.2.1 Guide samples. End item samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2.2 Standard samples. The finished cloth shall match the standard sample (see 6.4) for shade and appearance. Unless otherwise indicated, the finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 3.3.6.1.1).

3.3 Material. The cloth shall be such that the end item will conform to the Rules and Regulations of the Flammable Fabrics Act without the application of a flame retardant finish to the material or to the end item. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Jersey knit fleece cloth. The cloth for the body of the sweatpants shall be a jersey knit fleece cloth consisting of 58 ± 5 percent cotton, 1 to 2 percent rayon, and the remaining percentage acrylic fiber. The cloth shall be knit using the following facing, tie-in, and backing yarns (see 6.5.1):

Facing yarn	-	Blend of a minimum of 96 percent carded cotton with the remaining percentage rayon
Tie-in yarn	-	Unrelaxed high bulk acrylic or any acrylic yarn that will meet all specified requirements
Backing yarn	-	Blend of a minimum of 80 percent carded cotton with the remaining percentage acrylic fiber.

The cloth shall be napped on the back such that the degree of nap and appearance of nape approximate that of the standard sample. In the event that there is no standard sample, there shall be a "medium" degree of nap of sufficient density to prevent show through of backing and/or tie-in yarns. The finished cloth shall conform to the following requirements when tested as specified in 4.4.1.2.

Weight, ounces per square yard	8.2 ± 0.5
Wales per inch	21 (minimum)
Courses per inch	25 (minimum)
Bursting strength, pounds	75 (minimum)

3.3.2 Rib knit cloth. The cloth for the gusset shall be a plaited 1 x 1 rib knit cloth consisting of 47 ± 5 percent cotton, 2 to 3 percent rayon, and the remaining percentage acrylic fiber. The cloth shall be knit using the following facing and inside yarns (see 6.5.2):

Facing yarn	-	Blend of a minimum of 96 percent carded cotton with the remaining percentage rayon
Inside yarn	-	Unrelaxed high bulk acrylic or any acrylic yarn that will meet all specified requirements

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The finished cloth shall conform to the following requirements when tested as specified in 4.4.1.2.

Weight, ounces per square yard	8.5 ± 0.5
Wales per inch	18 (minimum)
Courses per inch	25 (minimum)
Bursting strength, pounds	90 (minimum)

3.3.3 Finish. No flame retardant finish or flame resistant finish shall be applied to the cloth or to the end item.

3.3.4 Color. All cloth for the sweatpants shall be Light Heather Gray 446.

3.3.4.1 Colorfastness. The finished cloth shall show fastness to light, laundering (after three cycles), and perspiration equal to or better than the standard sample, or equal to or better than the rating of "good" when tested as specified in 4.4.1.2. The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.4.1.2.

3.3.4.2 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and having a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.3.5 pH. The pH of the finished cloth shall be 5.0 to 8.5 when tested as specified in 4.4.1.2.

3.3.6 Webbing, elastic. The elastic webbing for the waist shall be woven using 12/2 ply cotton warp yarns (crochet) and an elastomer made of rubber wrapped with 3 ply cotton yarn and then 1 ply cotton yarn. The elastic webbing shall conform to the following requirements:

Width, inches	7/8 (± 1/16)
Weight, ounces per linear yard	0.046
Warp ends, full width (crochet)	24
Elastomer ends, full width	9
Filling yarns per inch	38 (± 4)
Breaking strength at maximum extension, pounds per foot	70 (minimum)

3.3.7 Thread. The thread for seaming and stitching shall be cotton-covered polyester conforming to MIL-T-43548 or spun polyester conforming to MIL-T-43624, dyed light gray DOD STD thread shade AX, C.A. 66064 and shall show colorfastness to laundering, perspiration, and light equal to or better than the standard sample or equal to or better than the rating of "good".

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3.3.8 Drawcord. The drawcord shall be 3/16 inch diameter 100 percent filament nylon, hot-knifed into lengths of 45 to 65 inches. The nylon shall be bright, regular or high tenacity polyamide having a 840 nominal denier and having a minimum breaking strength of 150 pounds. The drawcord shall be a basket weave with no core. The drawcord shall be light gray DOD Standard Shade AV, C.A. 66064 and shall show "good" colorfastness to laundering.

3.3.9 Eyelets. The eyelet for the waist drawcord shall be one piece, brass (BNE 133), conforming to MIL-E-20652/1. As an option, a washer equal to or better than Stimpson Part No. C-84 may be used with the eyelet.

3.3.10 Rubber band. The rubber band for the ankles shall be natural rubber approximately 1/4 inch wide, 4-1/2 inches long, and 3/64 inch thick.

3.3.11 Label. Each sweatpants shall have a combination size, identification and instruction label conforming to type VI, class 14 of DDD-L-20. All required information shall be printed on one side, or the size and identification information may be on one side and care instructions on the other side provided the care instructions are right side up when the label is turned over. All requirements including size of characters shall be as specified in DDD-L-20. The label shall show fastness to laundering. The body of the instruction label shall be as follows:

CARE INSTRUCTIONS

Hand or machine wash in cold or warm water (maximum 120°F).
Do not Bleach or Iron. Tumble dry at low or medium setting (maximum 140°F).

3.4 Design. The sweatpants shall be an all-around elastic waist design with drawcord, and elastic gathered ankles.

3.5 Construction. The construction shall conform in all respects to the requirements specified in table I and herein. Figure 1 is furnished solely for guidance and information. Should variation from the document appear in figure 1, the document shall govern.

3.5.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

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3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Types 504, 506, and 607 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.1.1.1a. and 3.5.1.1.1b. The gage of the overedge stitching shall be 3/16 ± 1/16 inch. The guide and knives for the overedge machine shall be set to trim only the ravelled edges of the fabric.

3.5.2 Thread breaks and ends of seams. Ends of all seams and stitching when not caught in other seams shall be backtacked not less than 1/4 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks shall be secured by stitching back of the break not less than 1/2 inch.

3.6 Preshrinking. The knitted cloth shall be preshrunk by a mechanical shrinkage process before being cut, such that the sweatpants will meet all shrinkage requirements specified in 3.9.

3.7 Manufacturing operations requirements. The sweatpants shall be manufactured in accordance with all operations specified in table I. The contractor is not required to follow the exact sequence of operations as long as all requirements are met.

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NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. The sweatpants shall be cut so that all the component parts of the fleece shall be cut widthwise in the direction of the wales. b. Cut all the component parts of the fleece cloth for the sweatpants from one piece of fabric. c. Cut the gusset from the same piece of rib knit cloth for the sizes indicated. The wales of the rib knit shall be cut lengthwise in the direction of the wales. The rib knit shall be of sufficient width to allow double thickness. d. Cut the elastic for the waist of the sweatpants so that the finished measurements of the relaxed and extended waist conform to the finished measurement requirements.						
2.	Replacement of defective components. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages as specified in 4.4.2 are excluded and replaced with nondefective and properly matched material.						

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Shade and size marking. The cut parts of the sweatpants shall be marked, ticketed or bundled to insure a uniform size and shade throughout. Any method of marking may be used except: a. Corrosive metal fastening devices. b. Sew on type tickets. c. Adhesive type tickets which discolor the material or leave traces of paper or adhesive on the material after removal of the ticket. d. Pin tickets.						
4.	Seam gusset and legs.						
	a. Seam gusset side pieces to center piece. Gussets are two ply.	504	SSa-1	10-14	50/2	50/2	50/2
	b. Join gusset to left back leg panel at the crotch of the sweatpants.	504	SSa-1	10-14	50/2	50/2	50/2
	c. Join front and back left inside leg seam from the bottom of the leg to the front seat seam including the stitching of the gusset to the left front crotch of the sweatpants.	504	SSa-1	10-14	50/2	50/2	50/2
	d. Join front seat seam from top of the waist to the right front inside leg seam, including the stitching of the gusset to the right front crotch of the sweatpants.	504	SSa-1	10-14	50/2	50/2	50/2

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Seam gusset and legs. (cont'd)						
	e. Join back seat seam, right back gusset to right crotch, and front and back right inside leg seam in one continuous operation from the bottom of the leg to the top of the waist.	504	SSa-1	10-14	50/2	50/2	50/2
	f. Turn sweatpants right side out.						
	g. Coverseam stitch all seams including gusset seams.	406	SSh-2	8-12	50/2	50/2	50/2
5.	Seam elastic waist.						
	a. Seam ends of waist elastic together.	504	SSa-1	10-14	50/2	50/2	50/2
	b. Align elastic to the edge of pants waist on inside and evenly distributed, seam together (catching label 1/2 to 3/4 inch to either side of center back seam, with the size visible).	504	SSa-1	10-14	50/2	50/2	50/2
	c. Attach eyelet as close as possible to the center front seam 5/8 inch from bottom stitch of waist hem and 5/8 plus 1/4 minus 1/8 inch from top of fold at waist. Measurements are taken to the center of the eyelet.						
	d. Insert drawcord thru eyelet and tie double knots in ends.						
	e. Fold waist edge with elastic over 1-1/4 ± 1/4 inch with cord in tunnel and seam.	406	EFa-1	8-12	50/2	50/2	50/2
	f. Make drawcord even, tack in place at center back seam.	Bartack		28 per bartack	50/2	50/2	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Seam elastic ankles. a. Place rubber band around leg, fold leg bottoms to the inside 1 inch with rubber band in tunnel and cover seam around leg allowing rubber band to move freely.	406	EFa-1	8-12	50/2	50/2	50/2
7.	Trim, inspect, bundle. a. Trim and remove all ends of thread of stitching inside and outside of the sweatpants. b. Remove all spots, stains, and all visible shade markings.						

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3.8 Finished measurements. The measurements of the finished sweatpants shall conform to the requirements specified in table II. All measurements shall be taken without stretching the material, except for the waist extended requirement.

TABLE II. Finished measurements (in inches)

	Extra Small	Small	Medium	Large	Extra Large	Tolerance
Waist relaxed A/	11-1/2	12-1/2	13-1/2	14-1/2	15-1/2	+1, -1/2
Waist (min. extended) B/	16-1/2	18-1/2	20-1/2	22-1/2	24-1/2	+any, -1/2
Inseam C/	31	32	33	34	35	+1, -1/2
Outseam D/	40-1/2	42	43-1/2	45	46-1/2	± 3/4
Front rise and gusset E/	14-1/2	15	15-1/2	16	16-1/2	± 1/2
Back rise F/	14-1/2	15	15-1/2	16	16-1/2	± 1/2
Leg gusset width G/	5-1/2	5-1/2	5-1/2	5-1/2	5-1/2	± 1/4
Leg gusset length H/	8-3/4	8-3/4	8-3/4	8-3/4	8-3/4	± 1/4
Thigh (at gusset) I/	8-1/2	8-1/2	8-1/2	8-1/2	8-1/2	± 1/2
Bottom leg hem J/	1	1	1	1	1	± 1/4
Crotch gusset width K/	3-1/4	3-1/4	3-1/4	3-1/4	3-1/4	± 1/2
Crotch gusset length L/	21-3/4	21-3/4	21-3/4	21-3/4	21-3/4	± 1/2

A/ WAIST RELAXED lay flat, measure side to side, relaxed.

B/ WAIST EXTENDED extend waist until body fabric is flat (but not stretched). Measure side to side. (on knit fabric the waist can extend more than the measurement).

C/ INSEAM measure seam from center of crotch to leg end.

D/ OUTSEAM top of sweatpants at waist to leg end, following natural shape of side seam.

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E & F/ FRONT RISE/BACK RISE measure from top of sweatpants at waist to fold in crotch gusset, do not stretch.

G/ LEG GUSSET WIDTH measure relaxed where two (2) gussets are joined.

H/ LEG GUSSET LENGTH center of seam where two (2) gussets are joined to the point.

I/ THIGH measure from leg gusset seam to fold of outseam parallel to the direction of the wales.

J/ LEG HEM width of hem including stitching.

K/ CROTCH GUSSET WIDTH measure relaxed at center.

L/ CROTCH GUSSET LENGTH front rise point to back rise point.

3.9 Dimensional stability. The finished sweatpants, after five launderings and dryings, shall not elongate more than 5.5 percent or shrink more than 8.0 percent lot average with no sample unit to exceed 10.0 percent for the outseam and shall not shrink more than 10.0 percent lot average with no sample unit to exceed 12.0 percent for the thigh, when tested as specified in 4.4.4.

3.10 Workmanship. The finished sweatpants shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain defective material.

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4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3, and tested as specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed below conform to the specified requirements.

Characteristic	Requirement paragraph
Cloth flammability	3.3
Yarns, fiber identification and content	3.3.1 and 3.3.2
Cloth fiber content	3.3.1 and 3.3.2
Webbing material identification, construction, fiber content, and breaking strength	3.3.6
Drawcord, material identification, denier and breaking strength	3.3.8

4.2.1.2 Component testing. The jersey knit fleece and rib knit cloths shall be tested for the characteristics listed in table III. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table III shall be followed. The sample unit shall be 1 yard full width. The lot shall be rejected if one or more sample units fail to meet any requirement specified. The sample size shall be in accordancew with the following:

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Lot size (yards)	Sample size (sample units)
Up to 22,000	2
22,001 and over	3

TABLE III. Cloth component tests

Characteristic	Requirement reference	Test method
Weight (oz./sq. yd.)	3.3.1 and 3.3.2	5041
Wales and courses per inch	3.3.1 and 3.3.2	5070
Bursting strength	3.3.1 and 3.3.2	5120
Colorfastness to:		
Light (after 80 standard fading hours)	3.3.4.1	5660
Laundering (after 3 cycles)	3.3.4.1	5614 1/
Perspiration	3.3.4.1	5680 1/
Crocking	3.3.4.1	5651 1/
pH	3.3.5	2811

1/ If material shade is obtained by 4 percent black solution-dyed rayon fiber mixed with 96 percent natural face cotton in the facing yarns, a certificate of compliance shall be submitted and will be accepted for the stated requirement.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of sweatpants. The sample unit shall be one sweatpants. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor
Material defects	Any hole, cut, tear, dropped stitches or run, uneven napping, or mend 1/	101	
	Uneven, ridgy or cockled cloth 1/		201
	Fleeced cloth not napped on inside	102	
	Any slub (more than three times normal diameter of yarn and clearly visible) on outside of sweatpants 1/		202
	Snag 1/		203
	Permanent crease in fabric	103	

1/ For knitted fabric imperfection definitions, see MIL-STD-1491.

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Color and shade	Color not as specified	104	
	Variation in shade within a part or between two or more parts		204
	Streaks clearly visible		205
Cleanness	Spot or stain, clearly noticeable:		
	On outside:		
	-more than 1/4 inch	105	
	-up to 1/4 inch inclusive		206
	On inside		207
	Rancid odor	106	
	Any marking clearly visible on outside	107	
	One or more marking tickets not removed		208
	Thread ends not trimmed, or loose threads not removed from sweatpants		209
Component and assembly	Any component part or required operation omitted or added (unless otherwise classified herein)	108	
	Any operation not performed as specified (unless otherwise classified herein)		210
	Gussets not double thickness of rib knit cloth	109	
	Any component part not as specified	110	
Cutting	Any component part not cut in accordance with directional lines as indicated on pattern or in accordance with specification requirements	111	
Seams and stitching	Any open seam (except on hems)	112	
	Open seam on hems:		
	-1/4 to 1 inch		211
	-over 1 inch	113	
	NOTE: A seam shall be considered as being open when one or more stitches joining a seam are broken, or when two or more successive skipped stitches occur.		

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching (cont'd)	Seam or stitch type not as specified	114	
	Parts of sweatpants badly pleated, caught or twisted in any type of stitching	115	
	Seam twisted or puckered		212
	Thread breaks (all stitch types) backstitched less than 1/2 inch		213
	Ends of all seams and stitchings produced with 301 stitch type not caught in other seams or stitchings, or backtacked less than 1/2 inch		214
	Ends of a continuous line of stitching overlapping less than 1/2 inch		215
	Stitch tension:		
	-loose tension resulting in a loose seam	116	
	-tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	117	
	Raw edges:		
	-any on outside	118	
	-on inside, 1/8 to 1/2 inch (inclusive)		216
	-on inside, over 1/2 inch	119	
	Runoff, or any line of stitching not catching all plies of material to be seamed:		
	-up to 1/2 inch (inclusive)		217
	-more than 1/2 inch	120	
Label	Omitted, incorrect, or illegible, misplaced or not attached as specified	121	
Waist and ankle	Not smooth, has pleats, is twisted or is stretched	122	
	Not specified width	123	
Eyelet	Omitted or not tightly clinched	124	
Drawcord	Not tacked, or double knotted, ends not hot knifed		218

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4.4.3 End item dimensional examination. The end items shall be examined for conformance to the measurements specified in table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of sweatpants. The sample unit shall be one sweatpants. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Dimensional stability test. The end items shall be tested for shrinkage as specified in Method 105-1987 of AATCC using normal machine cycle 120° ± 5°F washing temperature, and delicate dryer setting (140°F). The sweatpants shall not be moistened or pressed before measuring. Measure-ments shall be taken to the nearest 0.1 inch with the sweatpants laid flat. Elongation shall be reported with a minus (-) sign. The lot size shall be expressed in sweatpants. The sample unit shall be three sweatpants from each size in the lot. The sample size shall be as indicated in 4.4.1.2. The lot shall be rejected if the elongation or shrinkage requirements specified in 3.9 are not met.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container. Open and non-continuous heat sealed seams and closure of polyethylene bag, omission of vent hole in poly-ethylene bag.
Content	Number of unit packs per shipping container is more or less than specified.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each of the sweatpants shall be neatly folded to approximate 14 by 11 inches and inserted in a close-fitting clear polyethylene bag. The bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The bag may be fabricated from polyethylene film tube or sheeting. The final closure of the bag may be heat sealed or a reverse flap or tuck. A 1/4 inch diameter hole shall be made at one corner of the bag to allow air to escape.

5.1.2 Commercial. The sweatpants shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Twenty-four sweatpants of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The inside dimensions of each shipping container shall approximate 24-1/2 inches in length, 16-1/2 inches in width, and 12 inches in depth when packing sizes X-small through large and 24-1/2 inches in length, 16-1/2 inches in width, and 19-1/2 inches in depth when packing size extra large. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to non-metallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Twenty-four sweatpants of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, minimum grade 200 of PPP-B-636. The inside dimensions of each shipping container shall approximate 24-1/2 inches in length, 16-1/2 inches in width and 12 inches in depth when packing sizes X-small through large and 24-1/2 inches in length, 16-1/2 inches in width, and 19-1/2 inches in depth when packing size extra large. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Sweatpants of one size only, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), sweatpants packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

Stock number
Nomenclature
Size
Quantity

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sweatpants are intended for use as a component of the Physical Fitness Uniform.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article sample is required (see 3.1, 4.3, and 6.3).
e. Level of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2).
h. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to end item samples, and/or standard samples, address the contracting activity issuing the invitation for bids.

6.5 Knitting cloth.

6.5.1 Jersey knit fleece cloth. A fabric meeting the requirements of this specification was knitted with the following yarns:

Facing: 28/1 cotton count-96/4 carded cotton/solution dyed black rayon.

Tie-In: 1/40 worsted count-unrelaxed Orlon, Type 42.

Backing: 14/1 cotton count-80/20 cotton/acrylic.

6.5.2 Rib knit cloth. A fabric meeting the requirements of this specification was knitted with the following yarns:

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Facing yarn: 28/1 cotton count-96/4 carded cotton/solution dyed black rayon.

Inside yarn: 1/40 worsted count-unrelaxed Orlon, Type 42.

6.6 Source of supply. A source of supply for the eyelets is:

Stimpson Co., Inc.
900 Sylvan AVenue
Bayport, NY 11705-1097

6.7 Subject term (key word) listing.

Athletic clothing
Knitted cloth

6.8 Changes from previous issue. Marginal notations are not used in this revision to dentify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - GL

Review activities:
Army - MD
DLA - CT

Preparing activity:
Army - GL

(Project 8415-A687)

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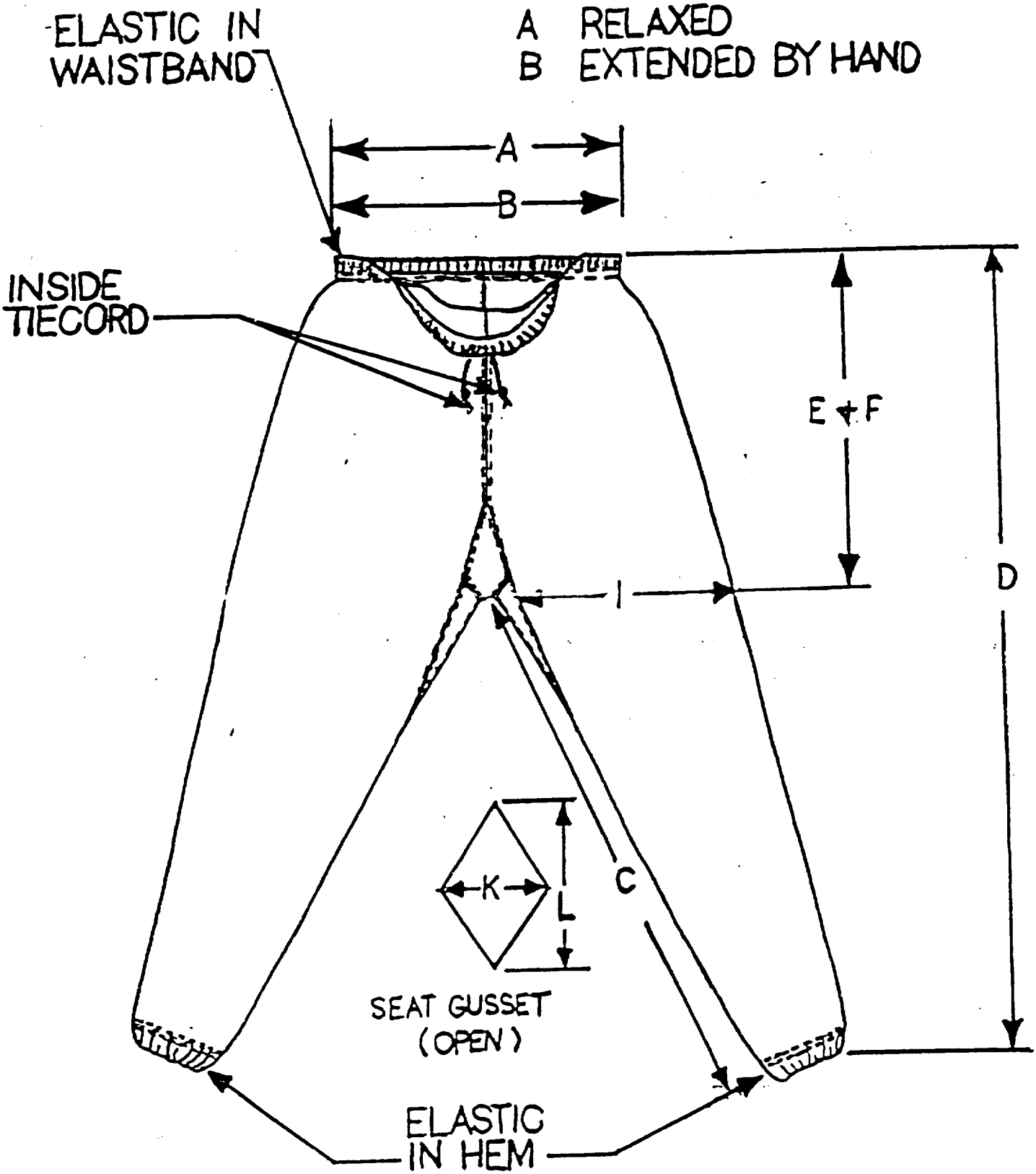


FIG.1 SWEATPANTS
PHYSICAL FITNESS UNIFORM

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